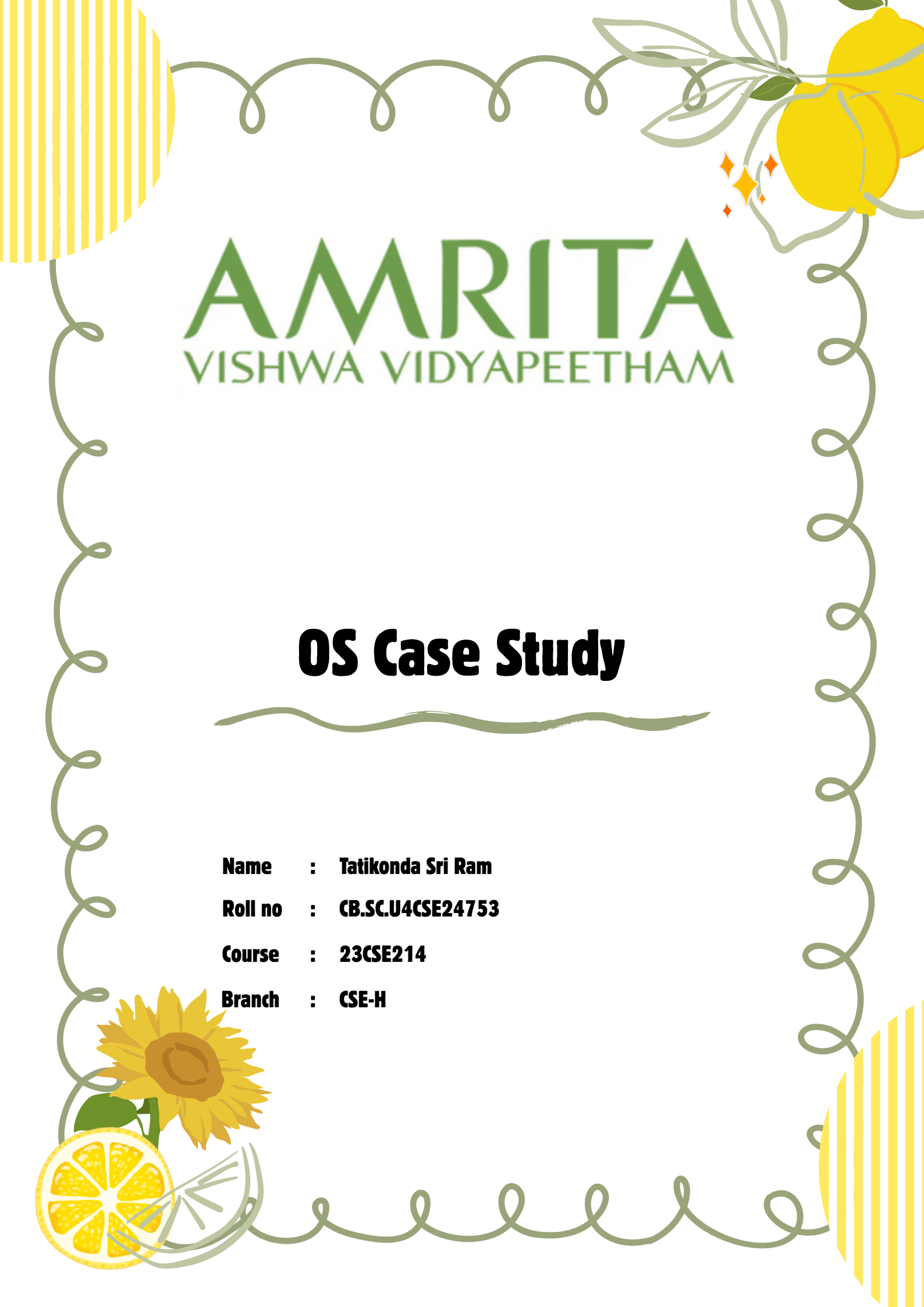




AMRITA

VISHWA VIDYAPEETHAM

OS Case Study

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Roll no : CB.SC.U4CSE24753

Course : 23CSE214

Branch : CSE-H

Stage 2:

```
ram@DESKTOP-R6JPL59:~/u4cse24753/xfs-interface$ ./xfs-interface
Unix-XFS Interace Version 2.0.
Type "help" for getting a list of commands.
# fdisk
Formatting Complete. "disk.xfs" created.
# exit
ram@DESKTOP-R6JPL59:~/u4cse24753/xfs-interface$ ./xfs-interface
Unix-XFS Interace Version 2.0.
Type "help" for getting a list of commands.
# df
0          -          1
1          -          1
2          -          1
3          -          1
4          -          1
5          -          1
6          -          1
7          -          1
8          -          1
9          -          1
10         -          1
11         -          1
12         -          1
13         -          1
14         -          1
15         -          1
16         -          1
17         -          1
18         -          1
19         -          1
20         -          1
21         -          1
22         -          1
23         -          1
24         -          1
25         -          1
26         -          1
27         -          1
28         -          1
29         -          1
30         -          1
31         -          1
32         -          1
33         -          1
34         -          1
35         -          1
36         -          1
37         -          1
```

38	-	1
39	-	1
40	-	1
41	-	1
42	-	1
43	-	1
44	-	1
45	-	1
46	-	1
47	-	1
48	-	1
49	-	1
50	-	1
51	-	1
52	-	1
53	-	1
54	-	1
55	-	1
56	-	1
57	-	1
58	-	1
59	-	1
60	-	1
61	-	1
62	-	1
63	-	1
64	-	1
65	-	1
66	-	1
67	-	1
68	-	1
69	-	0
70	-	0
71	-	0
72	-	0
73	-	0
74	-	0
75	-	0
76	-	0
77	-	0
78	-	0
79	-	0
80	-	0
81	-	0
82	-	0
83	-	0
84	-	0

85	-	0
86	-	0
87	-	0
88	-	0
89	-	0
90	-	0
91	-	0
92	-	0
93	-	0
94	-	0
95	-	0
96	-	0
97	-	0
98	-	0
99	-	0
100	-	0
101	-	0
102	-	0
103	-	0
104	-	0
105	-	0
106	-	0
107	-	0
108	-	0
109	-	0
110	-	0
111	-	0
112	-	0
113	-	0
114	-	0
115	-	0
116	-	0
117	-	0
118	-	0
119	-	0
120	-	0
121	-	0
122	-	0
123	-	0
124	-	0
125	-	0
126	-	0

127	-	0
128	-	0
129	-	0
130	-	0
131	-	0
132	-	0
133	-	0
134	-	0
135	-	0
136	-	0
137	-	0
138	-	0
139	-	0
140	-	0
141	-	0
142	-	0
143	-	0
144	-	0
145	-	0
146	-	0
147	-	0
148	-	0
149	-	0
150	-	0
151	-	0
152	-	0
153	-	0
154	-	0
155	-	0
156	-	0
157	-	0
158	-	0
159	-	0
160	-	0
161	-	0
162	-	0
163	-	0
164	-	0
165	-	0
166	-	0
167	-	0
168	-	0

169	-	0
170	-	0
171	-	0
172	-	0
173	-	0
174	-	0
175	-	0
176	-	0
177	-	0
178	-	0
179	-	0
180	-	0
181	-	0
182	-	0
183	-	0
184	-	0
185	-	0
186	-	0
187	-	0
188	-	0
189	-	0
190	-	0
191	-	0
192	-	0
193	-	0
194	-	0
195	-	0
196	-	0
197	-	0
198	-	0
199	-	0
200	-	0
201	-	0
202	-	0
203	-	0
204	-	0
205	-	0
206	-	0
207	-	0
208	-	0
209	-	0
210	-	0
211	-	0
212	-	0

```

479      -      0
480      -      0
481      -      0
482      -      0
483      -      0
484      -      0
485      -      0
486      -      0
487      -      0
488      -      0
489      -      0
490      -      0
491      -      0
492      -      0
493      -      0
494      -      0
495      -      0
496      -      0
497      -      0
498      -      0
499      -      0
500      -      0
501      -      0
502      -      0
503      -      0
504      -      0
505      -      0
506      -      0
507      -      0
508      -      0
509      -      0
510      -      0
511      -      0

No of Free Blocks = 443
Total no of Blocks = 512
# load --data $HOME/u4cse24753/sample.dat
# copy 3 4 $HOME/u4cse24753/inode_table.txt
# exit
ram@DESKTOP-R6JPL59:~/u4cse24753/xf-interface$ ./xf-interface
Unix-XFS Interface Version 2.0.
Type "help" for getting a list of commands.
# dump --inodeusertable
# copy 69 69 $HOME/u4cse24753/data.txt
# export sample.dat $HOME/u4cse24753/data.txt

```

```

ram@DESKTOP-R6JPL59: ~/u.  X  +  v
1  There is a plac
2  e where the sid
3  ewalk ends
4  And before the
5  street begins,
6  And there the g
7  rass grows soft
8  and white,
9  And there the s
10 un burns crimso
11 n bright,
12 And there the m
13 oon-bird rests
14 from his flight
15
16
17 To cool in the
18 peppermint wind
19 .
20

```

```

1  root
2  512
3  0
4  0
5  -1
6  -1
7  -1
8  5
9  -1
10 -1
11 -1
12 -1
13 -1
14 -1
15 -1
16 -1
17 2
18 sample.dat
19 18
20 1
21 1
22 -1
23 -1
24 -1
25 69
26 -1
27 -1
28 -1
29 -1
30 -1
31 -1
32 -1
33 -1
34 0
35 -1
36 -1
37 -1
38 -1
39 -1
40 -1
41 -1
42 -1
43 -1
44 -1

```

Assignment 1:

```
ram@DESKTOP-R6JPL59: ~/u. × + v
1 root
2 512
3 1
4 -1
5 -1
6 -1
7 -1
8 -1
9 sample.dat
10 18
11 2
12 -1
13 -1
14 -1
15 -1
16 -1
17 -1
18 0
19 -1
20 -1
21 -1
22 -1
23 -1
24 -1
25 -1
26 0
27 -1
28 -1
29 -1
30 -1
31 -1
32 -1
33 -1
34 0
35 -1
36 -1
37 -1
38 -1
39 -1
40 -1
41 -1
42 0
43 -1
44 -1
45 -1
46 1
```

Assignment 2:

ram@DESKTOP-R6JPL59: ~/u

1 1
2 root
3 512
4 0
5 0
6 -1
7 -1
8 -1
9 5
10 -1
11 -1
12 -1
13 -1
14 -1
15 -1
16 -1
17 2
18 sample.dat
19 18
20 1
21 1
22 -1
23 -1
24 -1
25 69
26 -1
27 -1
28 -1
29 -1
30 -1
31 -1
32 -1
33 -1
34 -1
35 0
36 -1
37 -1
38 -1
39 -1
40 -1
41 -1
42 -1
43 -1
44 -1
45 -1
46 -1

1 1
2 root
3 512
4 0
5 0
6 -1
7 -1
8 -1
9 5
10 -1
11 -1
12 -1
13 -1
14 -1
15 -1
16 -1
17 -1
18 -1
19 0
20 -1
21 -1
22 -1
23 -1
24 -1
25 -1
26 -1
27 -1
28 -1
29 -1
30 -1
31 -1
32 -1
33 -1
34 -1
35 0
36 -1
37 -1
38 -1
39 -1
40 -1
41 -1
42 -1
43 -1
44 -1
45 -1
46 -1

assignment2/inodeusertable_before.txttext1:1~/u4cse24753/stage2/assignment2/inodeusertable_after.txt1,3All

ram@DESKTOP-R6JPL59: ~/u

1 root
2 512
3 1
4 -1
5 -1
6 -1
7 -1
8 -1
9 sample.dat
10 18
11 2
12 -1
13 -1
14 -1
15 -1
16 -1
17 -1
18 0
19 -1
20 -1
21 -1
22 -1
23 -1
24 -1
25 -1
26 0
27 -1
28 -1
29 -1
30 -1
31 -1
32 -1
33 -1
34 0
35 -1
36 -1
37 -1
38 -1
39 -1
40 -1
41 -1
42 0
43 -1
44 -1
45 -1
46 -1

1 root
2 512
3 1
4 -1
5 -1
6 -1
7 -1
8 -1
9 -1
10 0
11 -1
12 -1
13 -1
14 -1
15 -1
16 -1
17 -1
18 0
19 -1
20 -1
21 -1
22 -1
23 -1
24 -1
25 -1
26 0
27 -1
28 -1
29 -1
30 -1
31 -1
32 -1
33 -1
34 0
35 -1
36 -1
37 -1
38 -1
39 -1
40 -1
41 -1
42 0
43 -1
44 -1
45 -1
46 -1

~/u4cse24753/stage2/assignment2/rootfile_before.txtassignment2/rootfile_after.txttext1:11,3All

Stage 3:

```
ram@DESKTOP-R6JPL59:~/u4cse24753$ cd xfs-interface/
ram@DESKTOP-R6JPL59:~/u4cse24753/xfs-interface$ ./xfs-interface load --os ../stage3
/helloworld.xsm
ram@DESKTOP-R6JPL59:~/u4cse24753/xfs-interface$ cd ..
ram@DESKTOP-R6JPL59:~/u4cse24753$ cd xsm/
ram@DESKTOP-R6JPL59:~/u4cse24753/xsm$ ./xsm
HELLO_WORLD
Machine is halting.
ram@DESKTOP-R6JPL59:~/u4cse24753/xsm$ █
```

Assignment 1:

```
MOV R0, 1
AGAIN:
MOV R2, 21
MOV R16, R0
PORT P1, R16
INR R0
OUT
GT R2, R0
JNZ R2, AGAIN
HALT
```

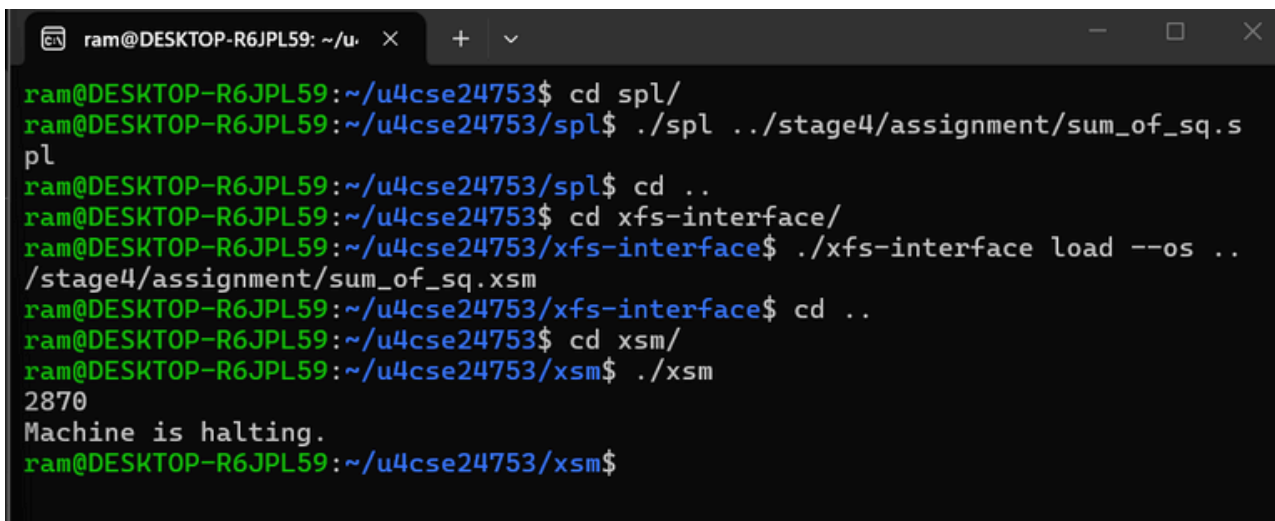
```
ram@DESKTOP-R6JPL59:~/u4cse24753/xfs-interface$ ./xfs-interface load --os ../stage3
/
assignment/      helloworld.xsm
ram@DESKTOP-R6JPL59:~/u4cse24753/xfs-interface$ ./xfs-interface load --os ../stage3
/assignment/new_boot.xsm
ram@DESKTOP-R6JPL59:~/u4cse24753/xfs-interface$ cd ..
ram@DESKTOP-R6JPL59:~/u4cse24753$ cd xsm/
ram@DESKTOP-R6JPL59:~/u4cse24753/xsm$ ./xsm
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
Machine is halting.
```


Stage 4:

```
ram@DESKTOP-R6JPL59:~/u4cse24753/stage4$ cd ..
ram@DESKTOP-R6JPL59:~/u4cse24753$ cp spl/spl_progs/oddnos.spl ./stage4/
ram@DESKTOP-R6JPL59:~/u4cse24753$ cd spl/
ram@DESKTOP-R6JPL59:~/u4cse24753/spl$ ./spl ../stage4/
assignment/ oddnos.spl
ram@DESKTOP-R6JPL59:~/u4cse24753/spl$ ./spl ../stage4/oddnos.spl
ram@DESKTOP-R6JPL59:~/u4cse24753/spl$ cd ..
ram@DESKTOP-R6JPL59:~/u4cse24753$ cd xfs-interface/
ram@DESKTOP-R6JPL59:~/u4cse24753/xfs-interface$ ./xfs-interface load --os
.gitignore          diskUtility.c      interface.h
AUTHORS             diskUtility.h      labels.c
LICENSE             exception.c        labels.h
Makefile            exception.h        memOrg.h
README.md           fileSystem.h       virtualDisk.c
constants.h         inode.c            virtualDisk.h
disk.c              inode.h            xfs-interface
disk.h              inodeusertable.txt
disk.xfs            interface.c
ram@DESKTOP-R6JPL59:~/u4cse24753/xfs-interface$ ./xfs-interface load --os ..
/stage4/
assignment/ oddnos.spl  oddnos.xsm
ram@DESKTOP-R6JPL59:~/u4cse24753/xfs-interface$ ./xfs-interface load --os ..
/stage4/
assignment/ oddnos.spl  oddnos.xsm
ram@DESKTOP-R6JPL59:~/u4cse24753/xfs-interface$ ./xfs-interface load --os ..
/stage4/oddnos.xsm
ram@DESKTOP-R6JPL59:~/u4cse24753/xfs-interface$ cd ..
ram@DESKTOP-R6JPL59:~/u4cse24753$ cd xsm/
ram@DESKTOP-R6JPL59:~/u4cse24753/xsm$ ./xsm
1
3
5
7
9
11
13
15
17
19
Machine is halting.
ram@DESKTOP-R6JPL59:~/u4cse24753/xsm$ |
```

Assignment 1:

```
alias counter R0;
alias temp R1;
counter = 1;
while (counter <= 20) do
    temp = temp + counter*counter;
    counter = counter + 1;
endwhile;
print temp;
```

A terminal window with a dark background and light green text. The window title is 'ram@DESKTOP-R6JPL59: ~/u'. The terminal shows the following commands and output:

```
ram@DESKTOP-R6JPL59:~/u4cse24753$ cd spl/
ram@DESKTOP-R6JPL59:~/u4cse24753/spl$ ./spl ../stage4/assignment/sum_of_sq.s
pl
ram@DESKTOP-R6JPL59:~/u4cse24753/spl$ cd ..
ram@DESKTOP-R6JPL59:~/u4cse24753$ cd xfs-interface/
ram@DESKTOP-R6JPL59:~/u4cse24753/xfs-interface$ ./xfs-interface load --os ..
/stage4/assignment/sum_of_sq.xsm
ram@DESKTOP-R6JPL59:~/u4cse24753/xfs-interface$ cd ..
ram@DESKTOP-R6JPL59:~/u4cse24753$ cd xsm/
ram@DESKTOP-R6JPL59:~/u4cse24753/xsm$ ./xsm
2870
Machine is halting.
ram@DESKTOP-R6JPL59:~/u4cse24753/xsm$
```

Stage 5:

```
ram@DESKTOP-R6JPL59:~/u4cse24753/xsm$ ./xsm --debug
Previous instruction at IP = 530: BRKP
Mode: KERNEL      PID: -1
Next instruction at IP = 532, Page No. = 1: JMP 534
debug> reg
R0: 1   R1:      R2:      R3:      R4:
R5:      R6:      R7:      R8:      R9:
R10:     R11:     R12:     R13:     R14:
R15:     R16: 1   R17: 0   R18:      R19:
P0:      P1:      P2:      P3:
BP:      SP:      IP: 532 PTBR:      PTLR:
EIP:     EC:      EPN:      EMA:
debug> s
Previous instruction at IP = 532: JMP 534
Mode: KERNEL      PID: -1
Next instruction at IP = 534, Page No. = 1: MOV R16,R0
debug> reg
R0: 1   R1:      R2:      R3:      R4:
R5:      R6:      R7:      R8:      R9:
R10:     R11:     R12:     R13:     R14:
R15:     R16: 1   R17: 0   R18:      R19:
P0:      P1:      P2:      P3:
BP:      SP:      IP: 534 PTBR:      PTLR:
EIP:     EC:      EPN:      EMA:
debug> s
Previous instruction at IP = 534: MOV R16,R0
Mode: KERNEL      PID: -1
Next instruction at IP = 536, Page No. = 1: ADD R16,1
debug> reg
R0: 1   R1:      R2:      R3:      R4:
R5:      R6:      R7:      R8:      R9:
R10:     R11:     R12:     R13:     R14:
R15:     R16: 1   R17: 0   R18:      R19:
P0:      P1:      P2:      P3:
BP:      SP:      IP: 536 PTBR:      PTLR:
EIP:     EC:      EPN:      EMA:
```

```

debug> s
Previous instruction at IP = 536: ADD R16,1
Mode: KERNEL      PID: -1
Next instruction at IP = 538, Page No. = 1: MOV R0,R16
debug> reg
R0: 1   R1:      R2:      R3:      R4:
R5:      R6:      R7:      R8:      R9:
R10:     R11:     R12:     R13:     R14:
R15:     R16: 2   R17: 0   R18:      R19:
P0:      P1:      P2:      P3:
BP:      SP:      IP: 538 PTBR:      PTLR:
EIP:     EC:      EPN:      EMA:

```

```

debug> c
Previous instruction at IP = 530: BRKP
Mode: KERNEL      PID: -1
Next instruction at IP = 532, Page No. = 1: JMP 534
debug> reg
R0: 3   R1:      R2:      R3:      R4:
R5:      R6:      R7:      R8:      R9:
R10:     R11:     R12:     R13:     R14:
R15:     R16: 1   R17: 0   R18:      R19:
P0:      P1:      P2:      P3:
BP:      SP:      IP: 532 PTBR:      PTLR:
EIP:     EC:      EPN:      EMA:

```

```

debug> c
Previous instruction at IP = 530: BRKP
Mode: KERNEL      PID: -1
Next instruction at IP = 532, Page No. = 1: JMP 534
debug> reg
R0: 5   R1:      R2:      R3:      R4:
R5:      R6:      R7:      R8:      R9:
R10:     R11:     R12:     R13:     R14:
R15:     R16: 1   R17: 0   R18:      R19:
P0:      P1:      P2:      P3:
BP:      SP:      IP: 532 PTBR:      PTLR:
EIP:     EC:      EPN:      EMA:

```

```

debug> c
Previous instruction at IP = 530: BRKP
Mode: KERNEL      PID: -1
Next instruction at IP = 532, Page No. = 1: JMP 534
debug> re
Unknown command "re". See "help" for more information.

```

```

debug> reg
R0: 7   R1:      R2:      R3:      R4:
R5:      R6:      R7:      R8:      R9:
R10:     R11:     R12:     R13:     R14:
R15:     R16: 1   R17: 0   R18:      R19:
P0:      P1:      P2:      P3:
BP:      SP:      IP: 532 PTBR:      PTLR:
EIP:     EC:      EPN:      EMA:

```

```

debug> c
Previous instruction at IP = 530: BRKP
Mode: KERNEL      PID: -1
Next instruction at IP = 532, Page No. = 1: JMP 534
debug> reg
R0: 9   R1:      R2:      R3:      R4:
R5:      R6:      R7:      R8:      R9:
R10:     R11:     R12:     R13:     R14:
R15:     R16: 1   R17: 0   R18:      R19:
P0:      P1:      P2:      P3:
BP:      SP:      IP: 532 PTBR:      PTLR:
EIP:     EC:      EPN:      EMA:

```

```

debug> c
Machine is halting.

```

```

ram@DESKTOP-R6JPL59:~/u4cse24753/xsm$ █

```



ram@DESKTOP-R6JPL59: ~/u



1 Page: 1

```
2 0: MOV R0,  
3 1: 0  
4 2: MOV R16,  
5 3: 10  
6 4: GE R16,  
7 5: R0  
8 6: JZ R16,  
9 7: 542  
10 8: MOV R16,  
11 9: R0  
12 10: MOD R16,  
13 11: 2  
14 12: MOV R17,  
15 13: 0  
16 14: NE R16,  
17 15: R17  
18 16: JZ R16,  
19 17: 534  
20 18: BRKP  
21 19: ·  
22 20: JMP 534  
23 21: ·  
24 22: MOV R16,  
25 23: R0  
26 24: ADD R16,  
27 25: 1  
28 26: MOV R0,  
29 27: R16  
30 28: JMP 514  
31 29: ·  
32 30: HALT  
33 31: ·  
34 32: ·
```

Stage 6:

```
ram@DESKTOP-R6JPL59:~/u4cse24753/xsm$ ./xsm --debug --timer 0
Previous instruction at IP = 12: BRKP
Mode: USER      PID: 0
Next instruction at IP = 14, Page No. = 0: ADD R0,1
debug> reg
R0: 1   R1: 1   R2: 1   R3:      R4:
R5:      R6:      R7:      R8:      R9:
R10:      R11:      R12:      R13:      R14:
R15:      R16: 1024      R17:      R18:      R19:
P0:      P1:      P2:      P3:
BP:      SP: 1023      IP: 14  PTBR: 29696      PTLR: 3
EIP:      EC:      EPN:      EMA:
debug> c
Previous instruction at IP = 12: BRKP
Mode: USER      PID: 0
Next instruction at IP = 14, Page No. = 0: ADD R0,1
debug> reg
R0: 2   R1: 4   R2: 1   R3:      R4:
R5:      R6:      R7:      R8:      R9:
R10:      R11:      R12:      R13:      R14:
R15:      R16: 1024      R17:      R18:      R19:
P0:      P1:      P2:      P3:
BP:      SP: 1023      IP: 14  PTBR: 29696      PTLR: 3
EIP:      EC:      EPN:      EMA:
debug> c
Previous instruction at IP = 12: BRKP
Mode: USER      PID: 0
Next instruction at IP = 14, Page No. = 0: ADD R0,1
debug> reg
R0: 3   R1: 9   R2: 1   R3:      R4:
R5:      R6:      R7:      R8:      R9:
R10:      R11:      R12:      R13:      R14:
R15:      R16: 1024      R17:      R18:      R19:
P0:      P1:      P2:      P3:
BP:      SP: 1023      IP: 14  PTBR: 29696      PTLR: 3
EIP:      EC:      EPN:      EMA:
debug> c
Previous instruction at IP = 12: BRKP
Mode: USER      PID: 0
Next instruction at IP = 14, Page No. = 0: ADD R0,1
```

```
debug> reg
R0: 4   R1: 16  R2: 1   R3:      R4:
R5:      R6:      R7:      R8:      R9:
R10:      R11:      R12:      R13:      R14:
R15:      R16: 1024      R17:      R18:      R19:
P0:      P1:      P2:      P3:
BP:      SP: 1023      IP: 14  PTBR: 29696      PTLR: 3
EIP:      EC:      EPN:      EMA:
debug> c
Previous instruction at IP = 12: BRKP
Mode: USER      PID: 0
Next instruction at IP = 14, Page No. = 0: ADD R0,1
debug> reg
R0: 5   R1: 25  R2: 1   R3:      R4:
R5:      R6:      R7:      R8:      R9:
R10:      R11:      R12:      R13:      R14:
R15:      R16: 1024      R17:      R18:      R19:
P0:      P1:      P2:      P3:
BP:      SP: 1023      IP: 14  PTBR: 29696      PTLR: 3
EIP:      EC:      EPN:      EMA:
debug> c
Machine is halting.
ram@DESKTOP-R6JPL59:~/u4cse24753/xsm$ █
```

Assignment 1:

INIT PROGRAM:

```
MOV R0, 1
MOV R2, 5
GE R2, R0
JZ R2, 2066
MOV R1, R0
MUL R1, R0
BRKP
ADD R0, 1
JMP 2050
INT 10
```

OS PROGRAM:

```
loadi(65,7);
loadi(66,8);
loadi(22,35);
loadi(23,36);
loadi(2,15);
loadi(3, 16);
PTBR = 29696;
PTLR = 9;
[PTBR+8] = 65;
[PTBR+9] = "0100";
[PTBR+10] = 66;
[PTBR+11] = "0100";
[PTBR+16] = 76;
[PTBR+17] = "0110";
[76*512] = 2048;
SP = 8*512;
ireturn;
```

```
ram@DESKTOP-R6JPL59:~/u4cse24753$ cd xfs-interface/
ram@DESKTOP-R6JPL59:~/u4cse24753/xfs-interface$ ./xfs-interface load --init ../stage6/
assignment/ haltprog.spl haltprog.xsm os_startup.spl os_startup.xsm squares.xsm
ram@DESKTOP-R6JPL59:~/u4cse24753/xfs-interface$ ./xfs-interface load --init ../stage6/assignment/squares.xsm
ram@DESKTOP-R6JPL59:~/u4cse24753/xfs-interface$ ./xfs-interface load --os ../stage6/assignment/os_startup.xsm
ram@DESKTOP-R6JPL59:~/u4cse24753/xfs-interface$ cd ..
ram@DESKTOP-R6JPL59:~/u4cse24753$ cd xsm/
ram@DESKTOP-R6JPL59:~/u4cse24753/xsm$ ./xsm
.gitignore debug.c disk.h exception.o machine.c main.o memory.o registers.o tokenize.c word.c
Makefile debug.h disk.o lex.yy.c machine.h mem parse.l simulator.c tokenize.h word.h
README.md debug.o exception.c lex.yy.o machine.o memory.c registers.c simulator.h tokenize.o word.o
constants.h disk.c exception.h lexer.h main.c memory.h registers.h simulator.o types.h xsm
ram@DESKTOP-R6JPL59:~/u4cse24753/xsm$ ./xsm --debug --timer 0
Previous instruction at IP = 2060: BRKP
Mode: USER PID: 0
Next instruction at IP = 2062, Page No. = 4: ADD R0,1
debug> reg
R0: 1 R1: 1 R2: 1 R3: R4:
R5: R6: R7: R8: R9:
R10: R11: R12: R13: R14:
R15: R16: 4096 R17: R18: R19:
P0: P1: P2: P3:
BP: SP: 4095 IP: 2062 PTBR: 29696 PTLR: 9
EIP: EC: EPN: EMA:
debug> c
Previous instruction at IP = 2060: BRKP
Mode: USER PID: 0
Next instruction at IP = 2062, Page No. = 4: ADD R0,1
```

```

Mode: USER      PID: 0
Next instruction at IP = 2062, Page No. = 4: ADD R0,1
debug> reg
R0: 2   R1: 4   R2: 1   R3:      R4:
R5:      R6:      R7:      R8:      R9:
R10:      R11:      R12:      R13:      R14:
R15:      R16: 4096      R17:      R18:      R19:
P0:      P1:      P2:      P3:
BP:      SP: 4095      IP: 2062      PTBR: 29696      PTLR: 9
EIP:      EC:      EPN:      EMA:
debug> c
Previous instruction at IP = 2060: BRKP
Mode: USER      PID: 0
Next instruction at IP = 2062, Page No. = 4: ADD R0,1
debug> reg
R0: 3   R1: 9   R2: 1   R3:      R4:
R5:      R6:      R7:      R8:      R9:
R10:      R11:      R12:      R13:      R14:
R15:      R16: 4096      R17:      R18:      R19:
P0:      P1:      P2:      P3:
BP:      SP: 4095      IP: 2062      PTBR: 29696      PTLR: 9
EIP:      EC:      EPN:      EMA:
debug> c
Previous instruction at IP = 2060: BRKP
Mode: USER      PID: 0
Next instruction at IP = 2062, Page No. = 4: ADD R0,1
debug> reg
R0: 4   R1: 16  R2: 1   R3:      R4:
R5:      R6:      R7:      R8:      R9:
R10:      R11:      R12:      R13:      R14:
R15:      R16: 4096      R17:      R18:      R19:
P0:      P1:      P2:      P3:
BP:      SP: 4095      IP: 2062      PTBR: 29696      PTLR: 9
EIP:      EC:      EPN:      EMA:
debug> c
Previous instruction at IP = 2060: BRKP
Mode: USER      PID: 0
Next instruction at IP = 2062, Page No. = 4: ADD R0,1
debug> reg
R0: 5   R1: 25  R2: 1   R3:      R4:
R5:      R6:      R7:      R8:      R9:
R10:      R11:      R12:      R13:      R14:
R15:      R16: 4096      R17:      R18:      R19:
P0:      P1:      P2:      P3:
BP:      SP: 4095      IP: 2062      PTBR: 29696      PTLR: 9
EIP:      EC:      EPN:      EMA:
debug> c
Machine is halting.
ram@DESKTOP-R6JPL59:~/u4cse24753/xsm$ █

```


Stage 7:

```
ram@DESKTOP-R6JPL59:~$ cd spl
-bash: cd: spl: No such file or directory
ram@DESKTOP-R6JPL59:~$ cd u4cse24753/
ram@DESKTOP-R6JPL59:~/u4cse24753$ ls
Makefile      inode_table.txt  stage10  stage5  stage9
data.txt      root_file.txt   stage2   stage6  xfs-interface
download.sh   sample.dat      stage3   stage7  xsm
expl          spl             stage4   stage8
ram@DESKTOP-R6JPL59:~/u4cse24753$ cd spl/
ram@DESKTOP-R6JPL59:~/u4cse24753/spl$ ./spl
.gitignore    file.o           spl.c
.temp         label.c          spl.h
AUTHORS       label.h          spl.o
LICENSE       label.o          spl_progs/
Makefile      lex.yy.c         splconstants.cfg
README.md     lex.yy.o         spllexer.l
data.c        node.c           splparser.y
data.h        node.h           y.tab.c
data.o        node.o           y.tab.h
file.c        samples/         y.tab.o
file.h        spl
ram@DESKTOP-R6JPL59:~/u4cse24753/spl$ ./spl ../stage7/
assignment/   os_startup.spl  squares.xsm
ram@DESKTOP-R6JPL59:~/u4cse24753/spl$ ./spl ../stage7/os_startup.spl
ram@DESKTOP-R6JPL59:~/u4cse24753/spl$ cd ..
ram@DESKTOP-R6JPL59:~/u4cse24753$ cd xfs-interface/
ram@DESKTOP-R6JPL59:~/u4cse24753/xfs-interface$ ./xfs-interface load --os ../stage7/os_startup.xsm
ram@DESKTOP-R6JPL59:~/u4cse24753/xfs-interface$ ./xfs-interface load --init ../stage7/squares.xsm
ram@DESKTOP-R6JPL59:~/u4cse24753/xfs-interface$ cd ..
ram@DESKTOP-R6JPL59:~/u4cse24753$ cd xsm/
ram@DESKTOP-R6JPL59:~/u4cse24753/xsm$ ./xsm --debug --timer 0
Previous instruction at IP = 2068: BRKP
Mode: USER      PID: 0
Next instruction at IP = 2070, Page No. = 4: ADD R0,1
```

```
debug> reg
R0: 1   R1: 1   R2: 1   R3:      R4:
R5:      R6:      R7:      R8:      R9:
R10:    R11:    R12:    R13:    R14:
R15:    R16: 4096   R17: 2056   R18:      R19:
P0:      P1:      P2:      P3:
BP:      SP: 4095   IP: 2070     PTBR: 29696   PTLR: 10
EIP:      EC:      EPN:      EMA:
debug> c
Previous instruction at IP = 2068: BRKP
Mode: USER      PID: 0
Next instruction at IP = 2070, Page No. = 4: ADD R0,1
debug> reg
R0: 2   R1: 4   R2: 1   R3:      R4:
R5:      R6:      R7:      R8:      R9:
R10:    R11:    R12:    R13:    R14:
R15:    R16: 4096   R17: 2056   R18:      R19:
P0:      P1:      P2:      P3:
BP:      SP: 4095   IP: 2070     PTBR: 29696   PTLR: 10
EIP:      EC:      EPN:      EMA:
debug> c
Previous instruction at IP = 2068: BRKP
Mode: USER      PID: 0
Next instruction at IP = 2070, Page No. = 4: ADD R0,1
debug> reg
R0: 3   R1: 9   R2: 1   R3:      R4:
R5:      R6:      R7:      R8:      R9:
R10:    R11:    R12:    R13:    R14:
R15:    R16: 4096   R17: 2056   R18:      R19:
P0:      P1:      P2:      P3:
BP:      SP: 4095   IP: 2070     PTBR: 29696   PTLR: 10
EIP:      EC:      EPN:      EMA:
debug> c
Previous instruction at IP = 2068: BRKP
Mode: USER      PID: 0
Next instruction at IP = 2070, Page No. = 4: ADD R0,1
```

```

debug> reg
R0: 4   R1: 16  R2: 1   R3:      R4:
R5:      R6:      R7:      R8:      R9:
R10:    R11:    R12:    R13:    R14:
R15:    R16: 4096   R17: 2056   R18:    R19:
P0:      P1:      P2:      P3:
BP:      SP: 4095   IP: 2070   PTBR: 29696   PTLR: 10
EIP:     EC:      EPN:     EMA:
debug> c
Previous instruction at IP = 2068: BRKP
Mode: USER      PID: 0
Next instruction at IP = 2070, Page No. = 4: ADD R0,1
debug> reg
R0: 5   R1: 25  R2: 1   R3:      R4:
R5:      R6:      R7:      R8:      R9:
R10:    R11:    R12:    R13:    R14:
R15:    R16: 4096   R17: 2056   R18:    R19:
P0:      P1:      P2:      P3:
BP:      SP: 4095   IP: 2070   PTBR: 29696   PTLR: 10
EIP:     EC:      EPN:     EMA:
debug> c
Machine is halting.
ram@DESKTOP-R6JPL59:~/u4cse24753/xsm$ █

```

Assignment 1:

OS PROGRAM:

```

loadi(63,13);
loadi(64,14);
loadi(65,7);
loadi(66,8);
loadi(22,35);
loadi(23,36);
loadi(2,15);
loadi(3, 16);
PTBR = PAGE_TABLE_BASE;
PTLR = 10;

```

```

//Library
[PTBR+0] = 63;
[PTBR+1] = "0100";
[PTBR+2] = 64;
[PTBR+3] = "0100";

```

```

//Heap
[PTBR+4] = 78;
[PTBR+5] = "0110";
[PTBR+6] = 79;
[PTBR+7] = "0110";

```

```
//Code
[PTBR+8] = 65;
[PTBR+9] = "0100";
[PTBR+10] = 66;
[PTBR+11] = "0100";
[PTBR+12] = -1;
[PTBR+13] = "0000";
[PTBR+14] = -1;
[PTBR+15] = "0000";

//Stack
[PTBR+16] = 76;
[PTBR+17] = "0110";
[PTBR+18] = 77;
[PTBR+19] = "0110";

[76*512] = [65*512+1];
SP = 8*512;
ireturn;
```

INIT PROGRAM:

```
0
2056
0
0
0
0
0
0
0
MOV R0,1
MOV R2, 5
GE R2, R0
JZ R2, 2076
MOV R1, R0
MUL R1, R0
MUL R1, R0
BRKP
ADD R0,1
JMP 2058
INT 10
```

```

ram@DESKTOP-R6JPL59:~/u4cse24753/spl$ ./spl ../stage7/assignment/os_startup.spl
ram@DESKTOP-R6JPL59:~/u4cse24753/spl$ cd ..
ram@DESKTOP-R6JPL59:~/u4cse24753$ cd xfs-interface/
ram@DESKTOP-R6JPL59:~/u4cse24753/xfs-interface$ ./xfs-interface load --os ../stage7/assignment/os_startup.xsm
ram@DESKTOP-R6JPL59:~/u4cse24753/xfs-interface$ ./xfs-interface load --init ../stage7/assignment/cubes.xsm
ram@DESKTOP-R6JPL59:~/u4cse24753/xfs-interface$ cd ..
ram@DESKTOP-R6JPL59:~/u4cse24753$ cd ./xsm/
ram@DESKTOP-R6JPL59:~/u4cse24753/xsm$ ./xsm --debug --timer 0
Previous instruction at IP = 2070: BRKP
Mode: USER      PID: 0
Next instruction at IP = 2072, Page No. = 4: ADD R0,1
debug> reg
R0: 1   R1: 1   R2: 1   R3:      R4:
R5:      R6:      R7:      R8:      R9:
R10:      R11:      R12:      R13:      R14:
R15:      R16: 4096      R17: 2056      R18:      R19:
P0:      P1:      P2:      P3:
BP:      SP: 4095      IP: 2072      PTBR: 29696      PTLR: 10
EIP:      EC:      EPN:      EMA:
debug> c
Previous instruction at IP = 2070: BRKP
Mode: USER      PID: 0
Next instruction at IP = 2072, Page No. = 4: ADD R0,1
debug> reg
R0: 2   R1: 8   R2: 1   R3:      R4:
R5:      R6:      R7:      R8:      R9:
R10:      R11:      R12:      R13:      R14:
R15:      R16: 4096      R17: 2056      R18:      R19:
P0:      P1:      P2:      P3:
BP:      SP: 4095      IP: 2072      PTBR: 29696      PTLR: 10
EIP:      EC:      EPN:      EMA:
debug> c
Previous instruction at IP = 2070: BRKP
Mode: USER      PID: 0
Next instruction at IP = 2072, Page No. = 4: ADD R0,1
debug> reg
R0: 3   R1: 27  R2: 1   R3:      R4:
R5:      R6:      R7:      R8:      R9:
R10:      R11:      R12:      R13:      R14:
R15:      R16: 4096      R17: 2056      R18:      R19:
P0:      P1:      P2:      P3:
BP:      SP: 4095      IP: 2072      PTBR: 29696      PTLR: 10
EIP:      EC:      EPN:      EMA:
debug> c
Previous instruction at IP = 2070: BRKP
Mode: USER      PID: 0
Next instruction at IP = 2072, Page No. = 4: ADD R0,1

```

```

Next instruction at IP = 2072, Page No. = 4: ADD R0,1
debug> reg
R0: 4   R1: 64  R2: 1   R3:      R4:
R5:      R6:      R7:      R8:      R9:
R10:      R11:      R12:      R13:      R14:
R15:      R16: 4096      R17: 2056      R18:      R19:
P0:      P1:      P2:      P3:
BP:      SP: 4095      IP: 2072      PTBR: 29696      PTLR: 10
EIP:      EC:      EPN:      EMA:
debug> c
Previous instruction at IP = 2070: BRKP
Mode: USER      PID: 0
Next instruction at IP = 2072, Page No. = 4: ADD R0,1
debug> reg
R0: 5   R1: 125 R2: 1   R3:      R4:
R5:      R6:      R7:      R8:      R9:
R10:      R11:      R12:      R13:      R14:
R15:      R16: 4096      R17: 2056      R18:      R19:
P0:      P1:      P2:      P3:
BP:      SP: 4095      IP: 2072      PTBR: 29696      PTLR: 10
EIP:      EC:      EPN:      EMA:
debug> c
Machine is halting.
ram@DESKTOP-R6JPL59:~/u4cse24753/xsm$ █

```

Stage 8:

[illegible]

Stage 9:

[illegible]

Assignment 1:

TIMER INTERRUPT:

```
[PROCESS_TABLE + ([SYSTEM_STATUS_TABLE + 1] * 16) + 13] = SP;  
SP = [PROCESS_TABLE + ([SYSTEM_STATUS_TABLE + 1] * 16) + 11] * 512 - 1;  
backup;  
print "TIMER";  
print [SYSTEM_STATUS_TABLE + 1];  
restore;  
SP = [PROCESS_TABLE + ([SYSTEM_STATUS_TABLE + 1] * 16) + 13];  
ireturn;
```

```
ram@DESKTOP-R6JPL59:~/u4cse24753$ cd spl/  
ram@DESKTOP-R6JPL59:~/u4cse24753/spl$ ./spl ../stage9/assignment/sample_timer.spl  
ram@DESKTOP-R6JPL59:~/u4cse24753/spl$ cd ..  
ram@DESKTOP-R6JPL59:~/u4cse24753$ cd xfs-interface/  
ram@DESKTOP-R6JPL59:~/u4cse24753/xfs-interface$ ./xfs-interface load --int=timer  
.gitignore      constants.h      diskUtility.h    inode.h          memOrg.h  
AUTHORS          disk.c           exception.c       interface.c       virtualDisk.c  
LICENSE          disk.h           exception.h       interface.h       virtualDisk.h  
Makefile         disk.xfs         fileSystem.h     labels.c         xfs-interface  
README.md        diskUtility.c    inode.c          labels.h  
ram@DESKTOP-R6JPL59:~/u4cse24753/xfs-interface$ ./xfs-interface load --int=timer ../stage9/assignment/sample_timer  
.xsm  
ram@DESKTOP-R6JPL59:~/u4cse24753/xfs-interface$ cd ..  
ram@DESKTOP-R6JPL59:~/u4cse24753$ cd ./x  
-bash: cd: ./x: No such file or directory  
ram@DESKTOP-R6JPL59:~/u4cse24753$ cd ./xsm/  
ram@DESKTOP-R6JPL59:~/u4cse24753/xsm$ ./xsm --timer 2  
TIMER  
0  
TIMER  
0  
TIMER  
0  
TIMER  
0  
TIMER  
0  
TIMER  
0  
TIMER  
0  
TIMER  
0  
TIMER  
0  
TIMER  
0  
TIMER  
0  
TIMER  
0  
TIMER  
0  
TIMER  
0  
TIMER  
0  
TIMER  
0  
TIMER  
0  
TIMER  
0  
TIMER  
0  
TIMER  
0  
TIMER  
0  
TIMER  
0  
TIMER  
0  
TIMER  
0  
TIMER  
0  
TIMER  
0  
TIMER  
0  
Machine is halting.  
ram@DESKTOP-R6JPL59:~/u4cse24753/xsm$ █
```

Stage 10:

```
ram@DESKTOP-R6JPL59:~/u4cse24753$ cd spl/
ram@DESKTOP-R6JPL59:~/u4cse24753/spl$ ./spl ../stage10/write_system_call.spl
ram@DESKTOP-R6JPL59:~/u4cse24753/spl$ ./spl ../spl_progs/sample_int7.spl
ram@DESKTOP-R6JPL59:~/u4cse24753/spl$ cd ..
ram@DESKTOP-R6JPL59:~/u4cse24753$ cd xfs-interface/
ram@DESKTOP-R6JPL59:~/u4cse24753/xfs-interface$ ./xfs-interface load --init ../stage10/squares.xsm
ram@DESKTOP-R6JPL59:~/u4cse24753/xfs-interface$ ./xfs-interface load --int=7 ../stage10/
os_startup.spl      squares.xsm      write_system_call.xsm
os_startup.xsm      write_system_call.spl
ram@DESKTOP-R6JPL59:~/u4cse24753/xfs-interface$ ./xfs-interface load --int=7 ../stage10/write_system_call.xsm
ram@DESKTOP-R6JPL59:~/u4cse24753/xfs-interface$ cd ..
ram@DESKTOP-R6JPL59:~/u4cse24753$ cd xsm/
ram@DESKTOP-R6JPL59:~/u4cse24753/xsm$ ./xsm --timer 0
1
4
9
16
25
Machine is halting.
ram@DESKTOP-R6JPL59:~/u4cse24753/xsm$ █
```

Assignment 1:

INIT PROGRAM:

```
0
2056
0
0
0
0
0
0
0
MOV R0, 1
MOV R2, 20
GE R2, R0
JZ R2, 2108
MOV R1, R0
PUSH R0
PUSH R1
PUSH R2
MOV R0, 5
MOV R2, -2
PUSH R0
PUSH R2
PUSH R1
PUSH R0
PUSH R0
```



```
INT 7
POP R0
POP R0
POP R0
POP R0
POP R0
POP R2
POP R1
POP R0
ADD R0,1
JMP 2058
INT 10
```

```
ram@DESKTOP-R6JPL59:~/u4cse24753/xsm$ ./xsm --timer 10
TIMER
0
1
TIMER
0
TIMER
0
TIMER
0
2
TIMER
0
TIMER
0
3
TIMER
0
TIMER
0
TIMER
0
4
TIMER
0
TIMER
0
5
TIMER
0
TIMER
0
TIMER
0
6
```

```
TIMER
0
TIMER
0
7
TIMER
0
TIMER
0
TIMER
0
8
TIMER
0
TIMER
0
9
TIMER
0
TIMER
0
TIMER
0
10
TIMER
0
TIMER
0
11
TIMER
0
TIMER
0
TIMER
0
12
TIMER
0
TIMER
0
13
TIMER
0
```

```
TIMER
0
TIMER
0
14
TIMER
0
TIMER
0
15
TIMER
0
TIMER
0
16
TIMER
0
TIMER
0
17
TIMER
0
TIMER
0
18
TIMER
0
19
TIMER
0
20
TIMER
0
Machine is halting.
```